Supplementary Information

Efficient Wide-Bandgap Perovskite Photovoltaics with Homogeneous Halogen-Phase Distribution

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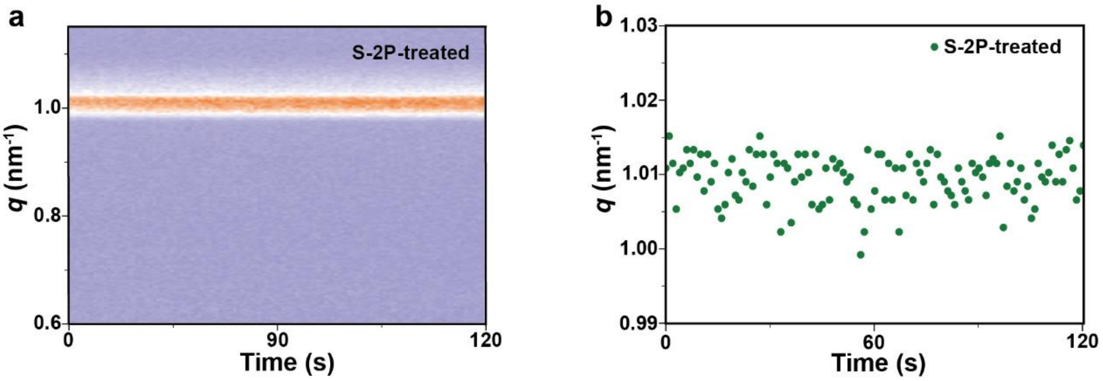
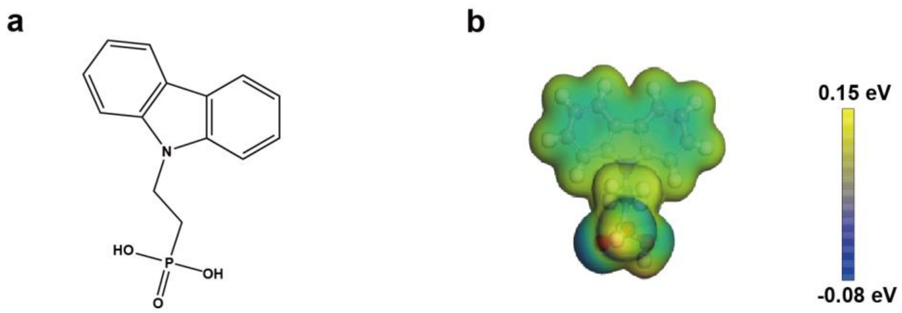
† These authors contributed equally.

Table of Contents

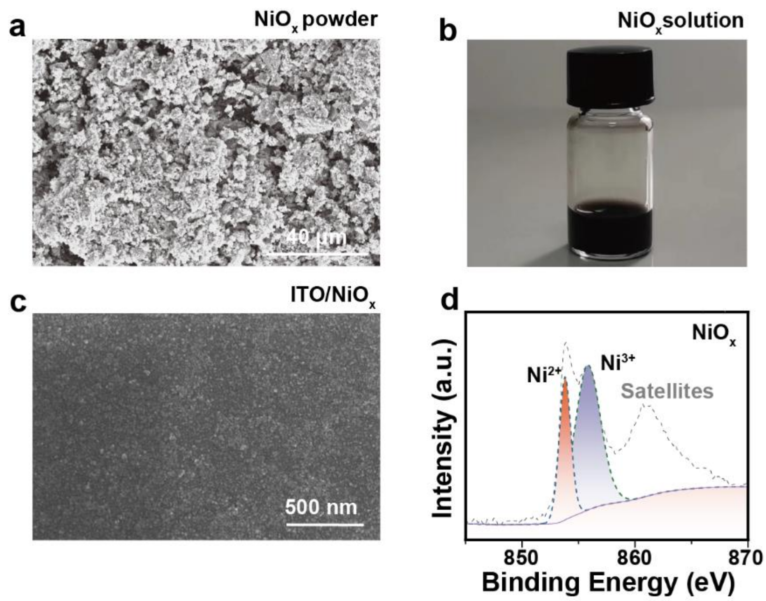
Supplementary Figures



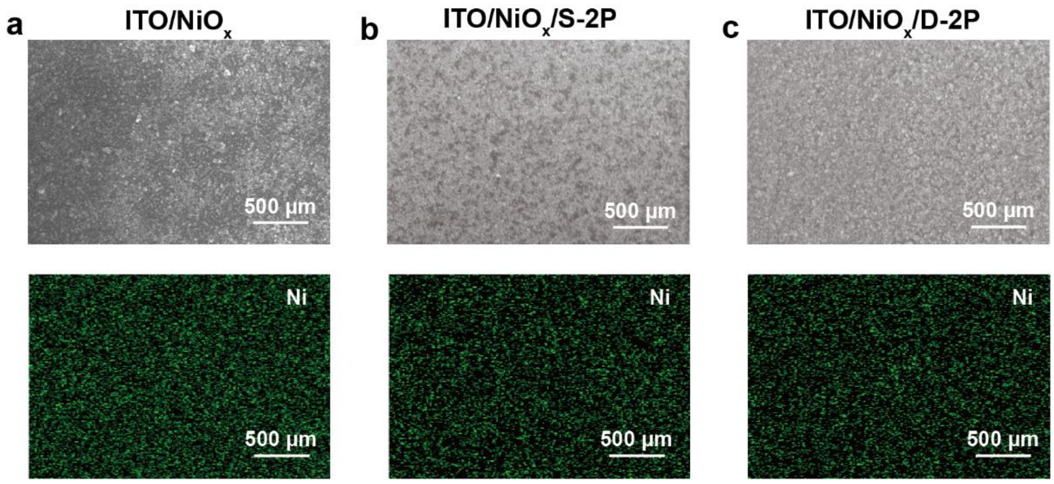
Supplementary Fig. 1 | a *In-situ* GIWAXS patterns of the S-2P-treated perovskite films at the annealing stage. **b** Evolution of q values over time.



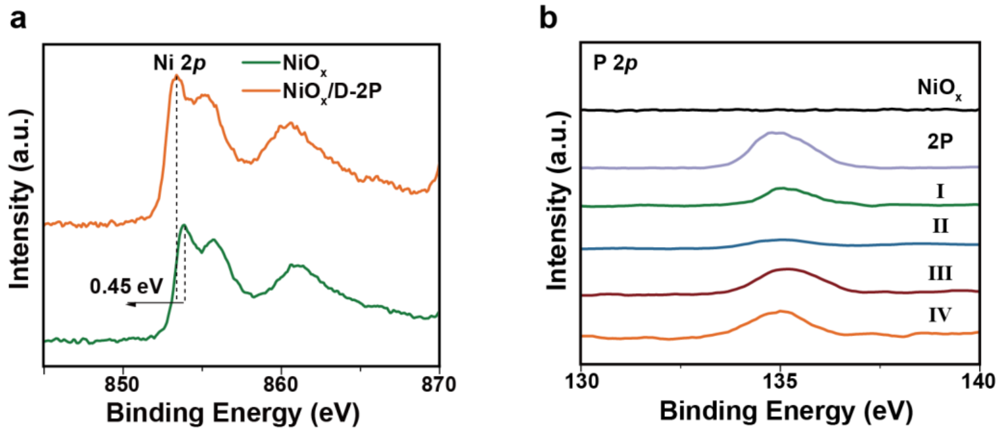
Supplementary Fig. 2 | a Structural formula and **b** Electrostatic potential of 2P molecule.



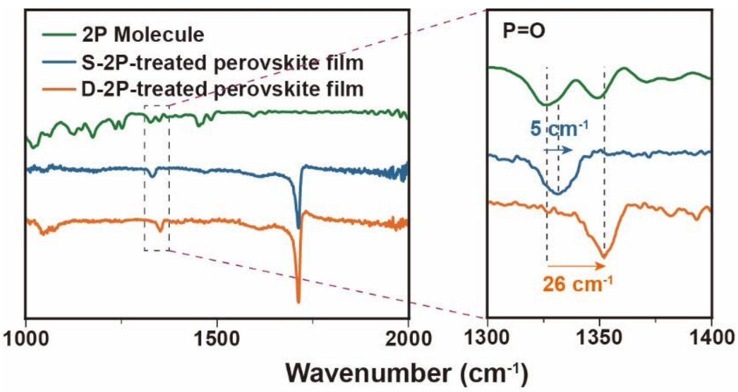
Supplementary Fig. 3 | a SEM images of NiO_x nanoparticles. **b** Optical photo of NiO_x solution. **c** SEM image and **d** XPS spectra of NiO_x film.



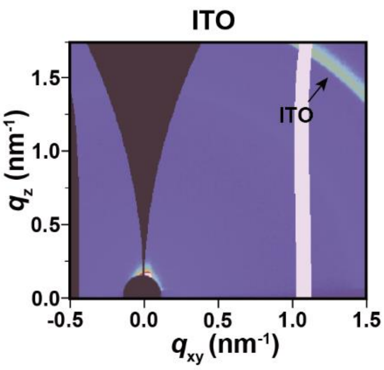
Supplementary Fig. 4 | SEM images and corresponding EDS mappings of **a** ITO/NiO$_x$, **b** ITO/NiO$_x$/S-2P, and **c** ITO/NiO$_x$/D-2P films.



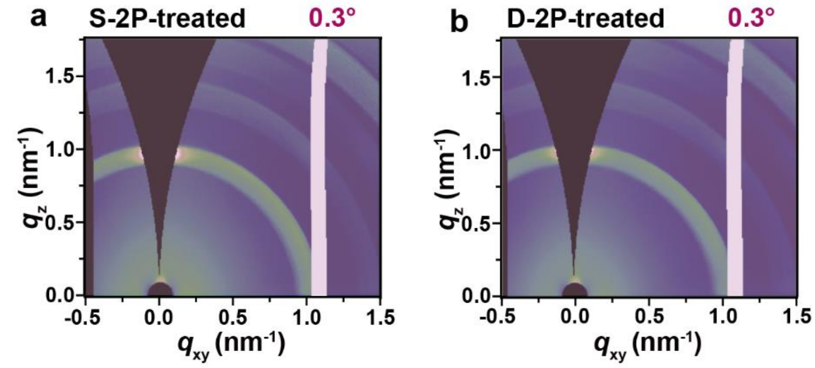
Supplementary Fig. 5 | a The XPS spectra of Ni 2*p* for NiO_x and NiO_x/2P films. **b** XPS spectra of P 2*p* for step I, II, III and IV.



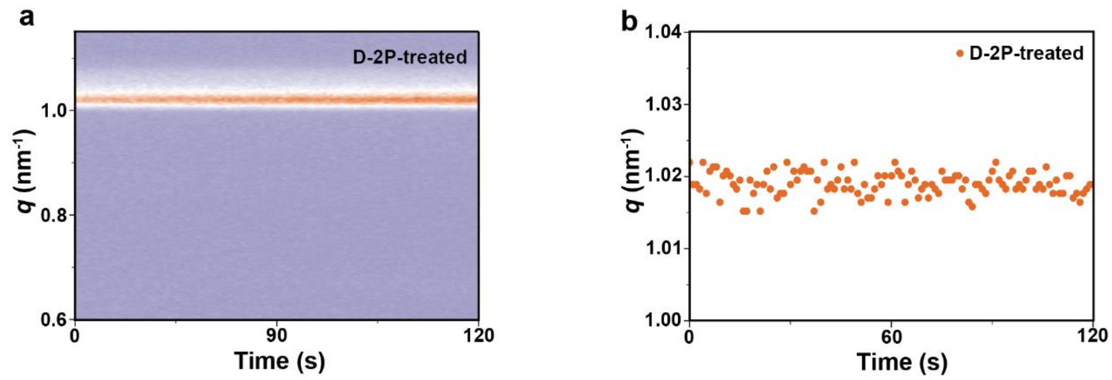
Supplementary Fig. 6 | FTIR of 2P molecule, S-2P-treated and D-2P-treated perovskite films.



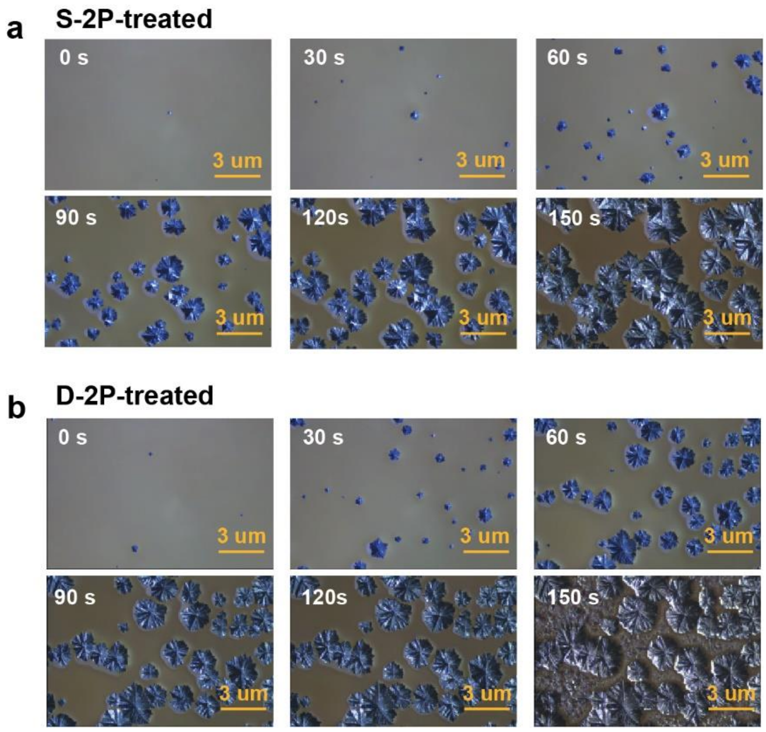
Supplementary Fig. 7 | GIWAXS pattern of ITO substrate.



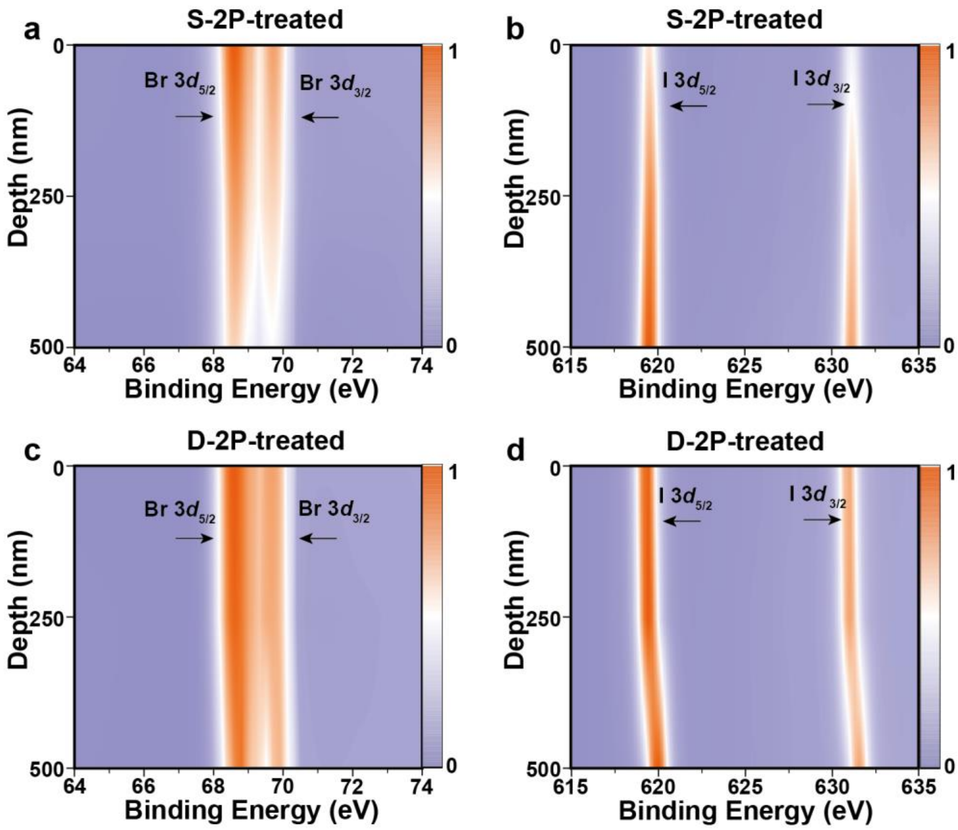
Supplementary Fig. 8 | GIWAXS for **a** S-2P-treated and **b** D-2P-treated perovskite films.



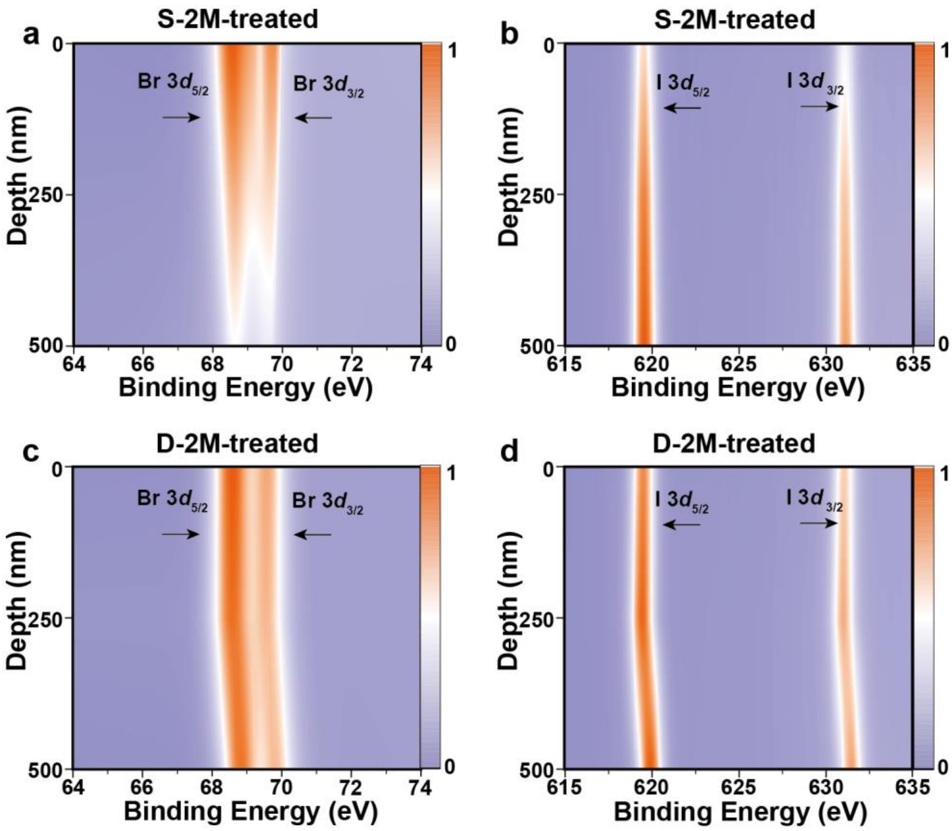
Supplementary Fig. 9 | a *In-situ* GIWAXS patterns of the D-2P-treated perovskite films at the annealing stage**. b** Evolution of q values over time.



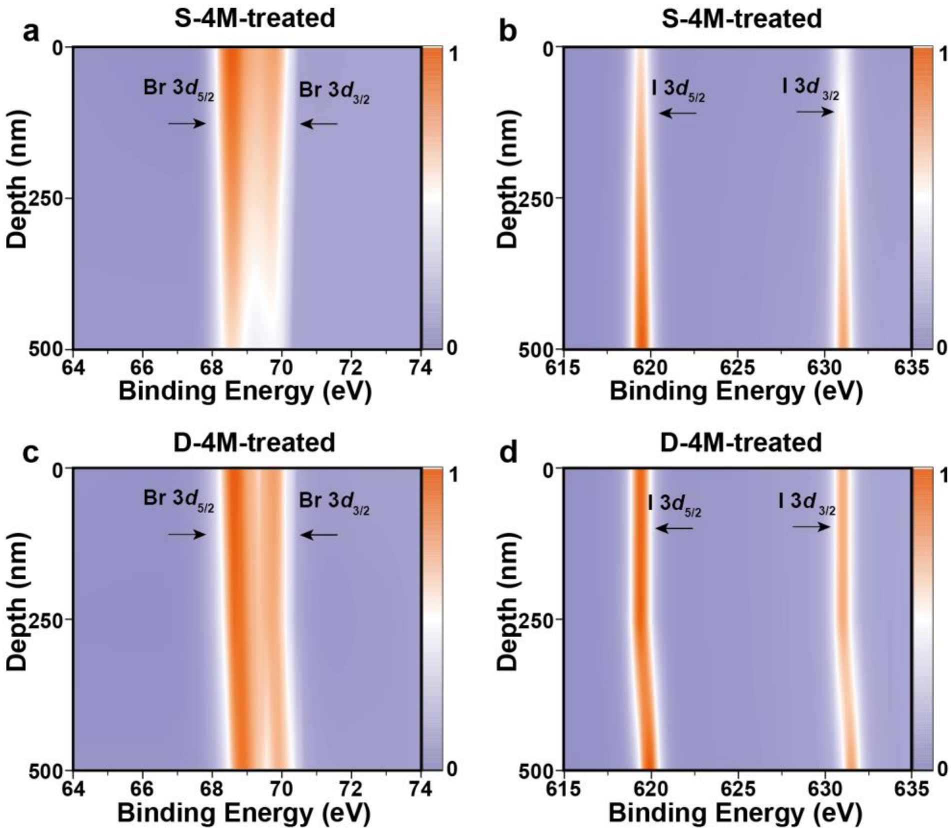
Supplementary Fig. 10 | Laser confocal microscope images of *in-situ* crystallization kinetics for **a** S-2P-treated and **b** D-2P-treated perovskite precursor solution.



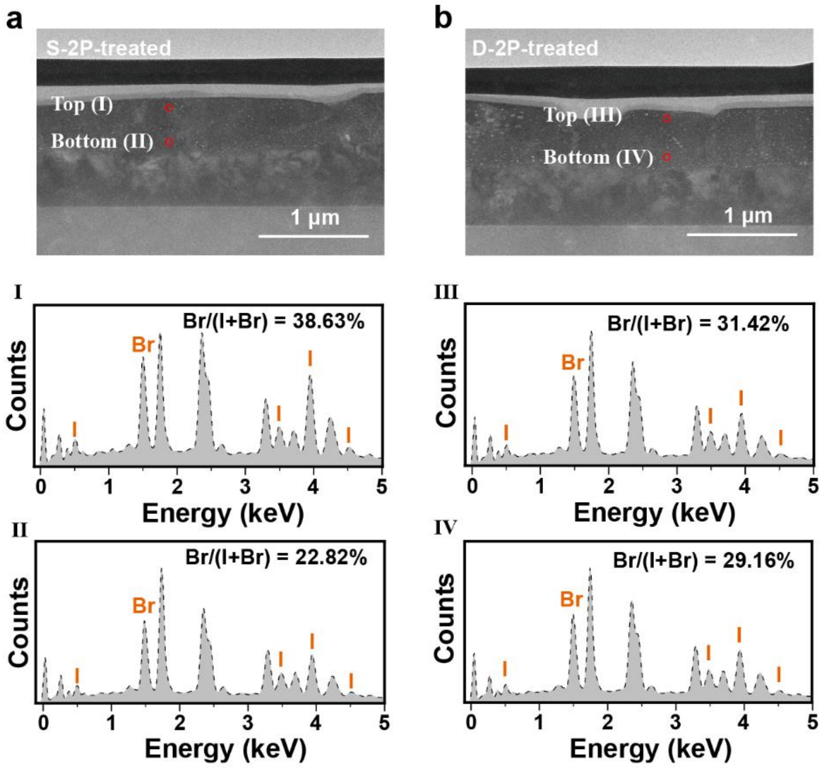
Supplementary Fig. 11 | Depth-profiling Br and I XPS spectra of **a, b** S-2P-treated and **c, d** D-2P-treated NiO_x-free perovskite films.



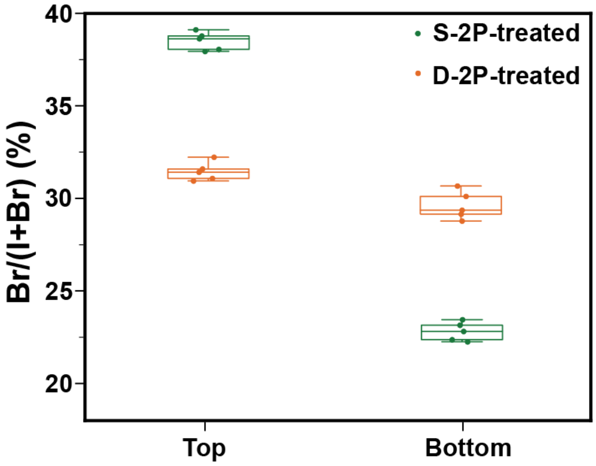
Supplementary Fig. 12 | Depth-profiling Br and I XPS spectra of **a, b** S-2M-treated and **c, d** D-2M-treated perovskite films.



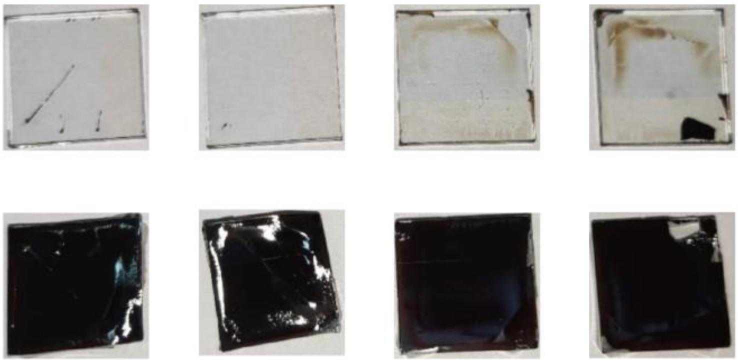
Supplementary Fig. 13 | Depth-profiling Br and I XPS spectra of **a, b** S-4M-treated and **c, d** D-4M-treated perovskite films.



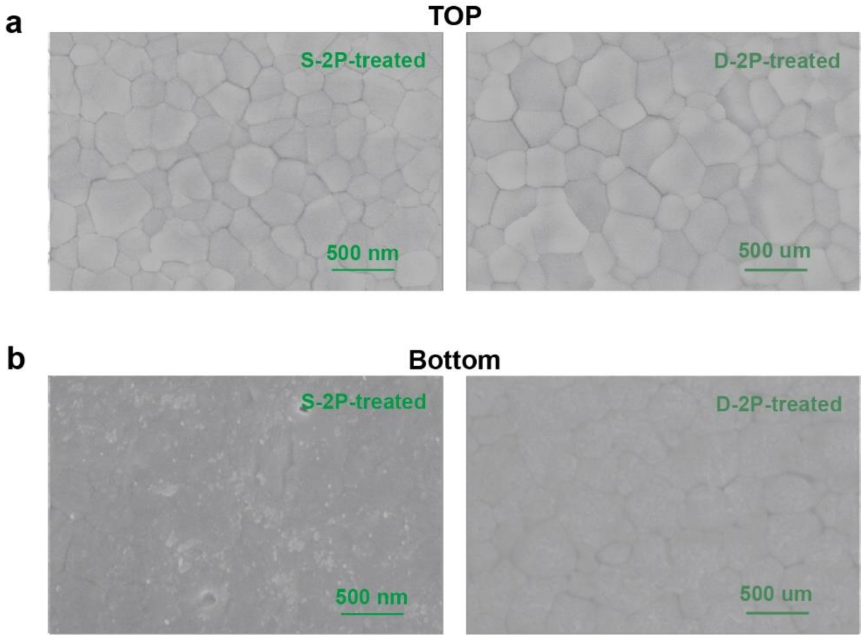
Supplementary Fig. 14 | The cross-section TEM images and corresponding EDS spectra for **a** S-2P-treated and **b** D-2P-treated perovskite films.



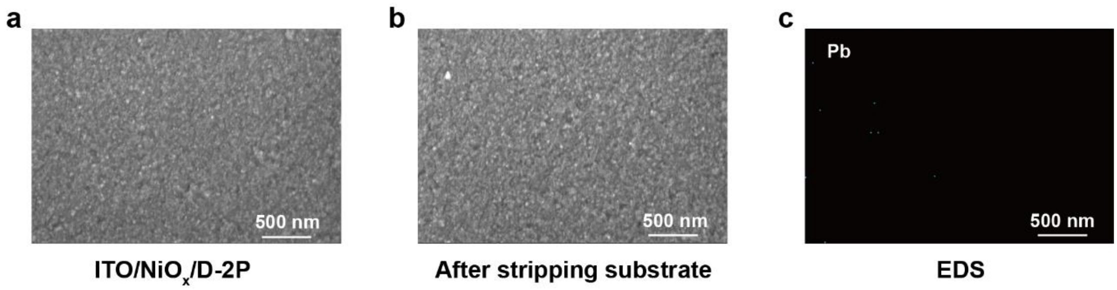
Supplementary Fig. 15 | The statistical halogen content of S-2P-treated and D-2P-treated perovskite films at different spatial positions.



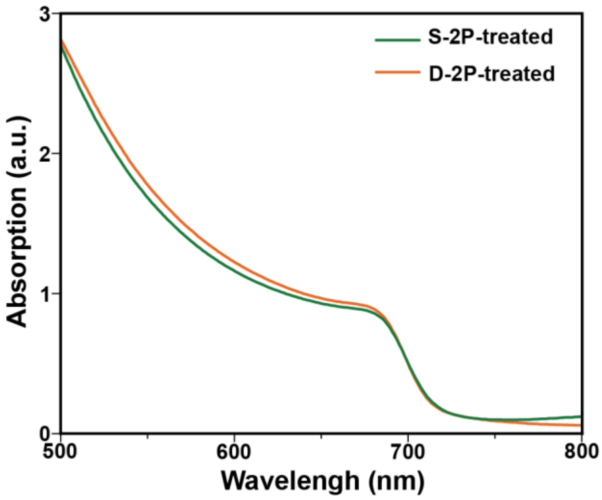
Supplementary Fig. 16 | Photos of perovskite films after stripping.



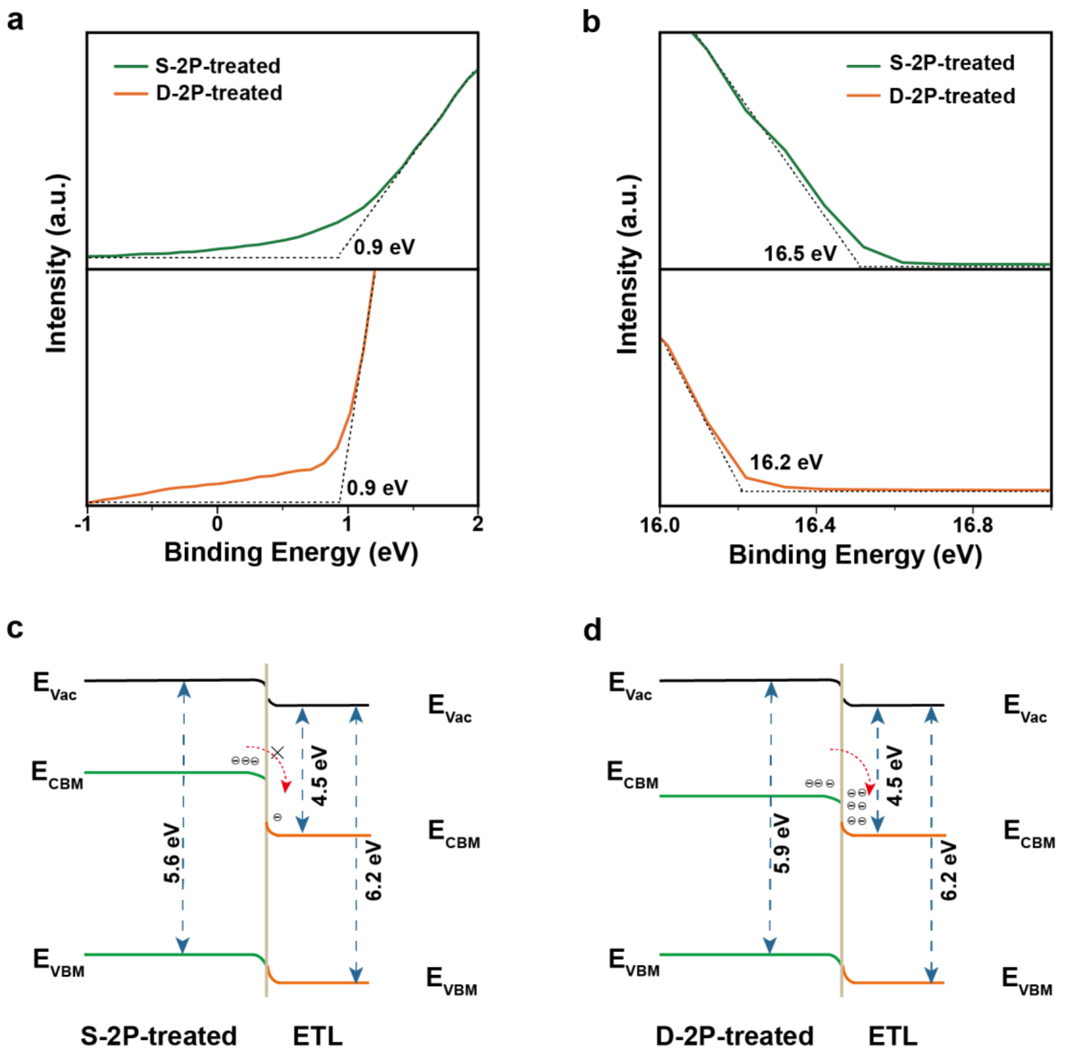
Supplementary Fig. 17 | SEM images of perovskite films for **a** TOP and **b** Bottom.



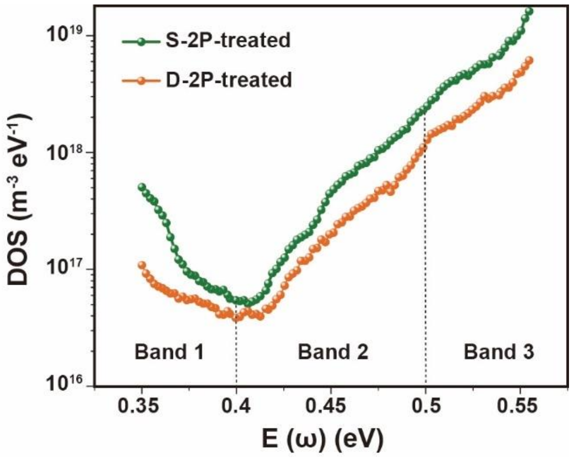
Supplementary Fig. 18 | a SEM image of ITO/NiO_x/D-2P film, **b** SEM image and **c** EDS mapping of Pb element of the substrate after stripping perovskite film.



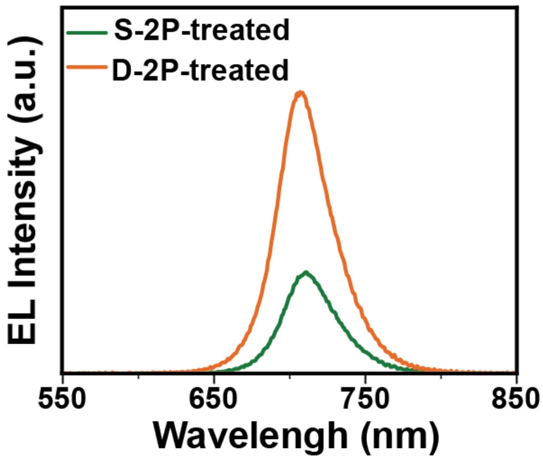
Supplementary Fig. 19 | UV-vis absorption spectra of the S-2P-treated and the D-2P-treated perovskite films.



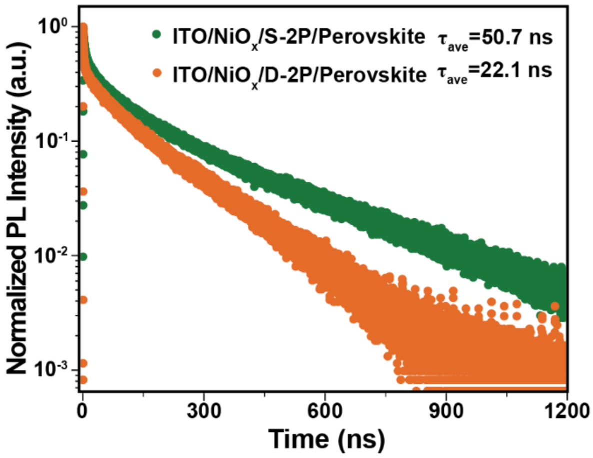
Supplementary Fig. 20 | a and **b** UPS spectra of perovskite films top. **c** and **d** Energy level diagram.



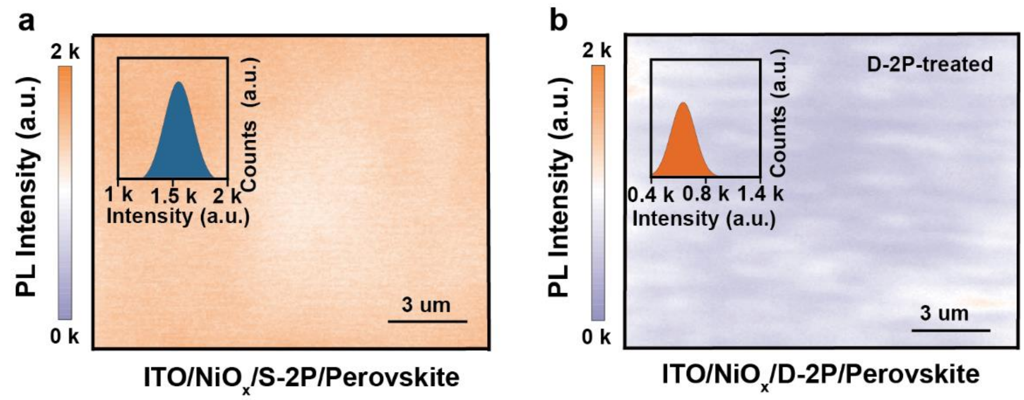
Supplementary Fig. 21 | TAS spectra of S-2P-treated and D-2P-treated perovskite films.



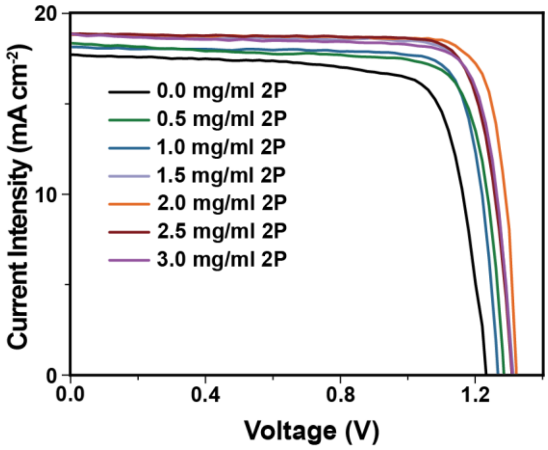
Supplementary Fig. 22 | EL spectra of S-2P-treated and D-2P-treated PSCs with bias voltage of 2 V.



Supplementary Fig. 23 | TRPL spectra of S-2P-treated and D-2P-treated perovskite films.



Supplementary Fig. 24 | PL intensity mapping for **a** S-2P-treated and **b** D-2P-treated perovskite films.



Supplementary Fig. 25 | *J-V* curves of different concentrations of 2P molecules.

Test and Calibration Center of New Energy Device and Module,
Shanghai Institute of Microsystem and Information Technology,
Chinese Academy of Sciences (SIMIT)

Measurement Report

Report No. 24TR031902

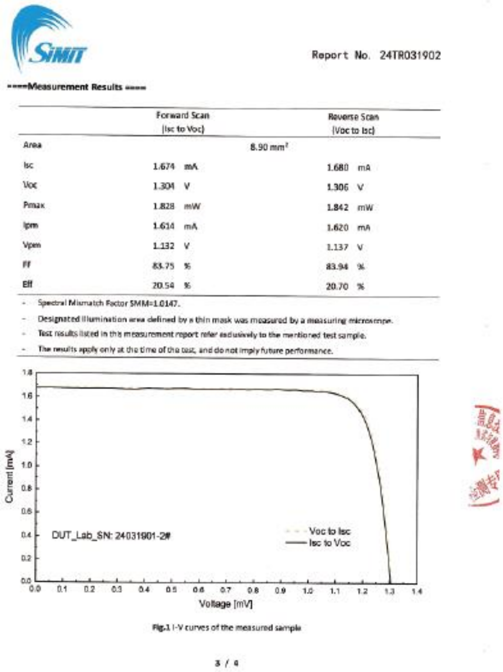
Client Name	Hebei University
Client Address	No. 180 Wusi Dong Road, Baoding, Hebei Province, China
Sample	Perovskite solar cell
Manufacturer	Hebei University
Measurement Date	19th March, 2024

Performed by:	Qiang Shi	Date:
Reviewed by:	Wenjie Zhao	Date:
Approved by:	Yucheng Liu	Date: 19/03/2024

Address: No.235 Chengbei Road, Jiading, Shanghai   Post Code:201800
E-mail: solarcell@mail.sim.ac.cn   Tel: +86-021-69976905

The measurement report without signature and seal are not valid.
This report shall not be reproduced, except in full, without the approval of SIMIT.

1 / 4



Report No. 24TR031902

====Measurement Results ====

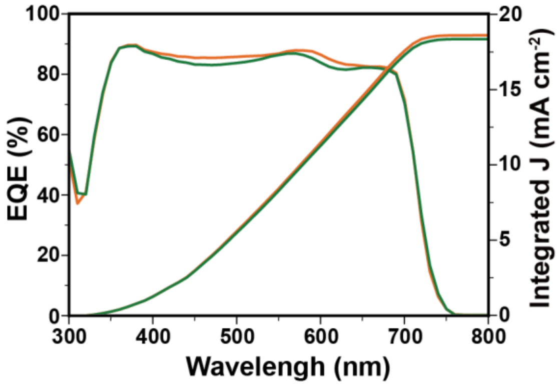
	Forward Scan (Isc to Voc)	Reverse Scan (Voc to Isc)
Area	8.90 mm^2	
Isc	1.674 mA	1.680 mA
Voc	1.304 V	1.306 V
Pmax	1.828 mW	1.842 mW
Ipm	1.614 mA	1.620 mA
Vpm	1.132 V	1.137 V
FF	83.75 %	83.94 %
Eff	20.54 %	20.70 %

- Spectral Mismatch Factor SMM=1.0147.
- Designated illumination area defined by a thin mask was measured by a measuring microscope.
- Test results listed in this measurement report refer exclusively to the mentioned test sample.
- The results apply only at the time of the test, and do not imply future performance.

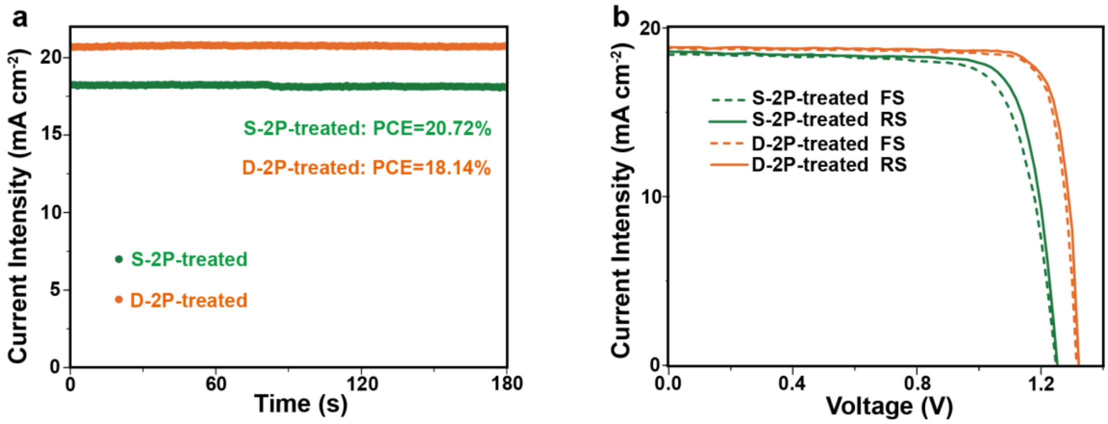
Fig.1 I-V curves of the measured sample

3 / 4

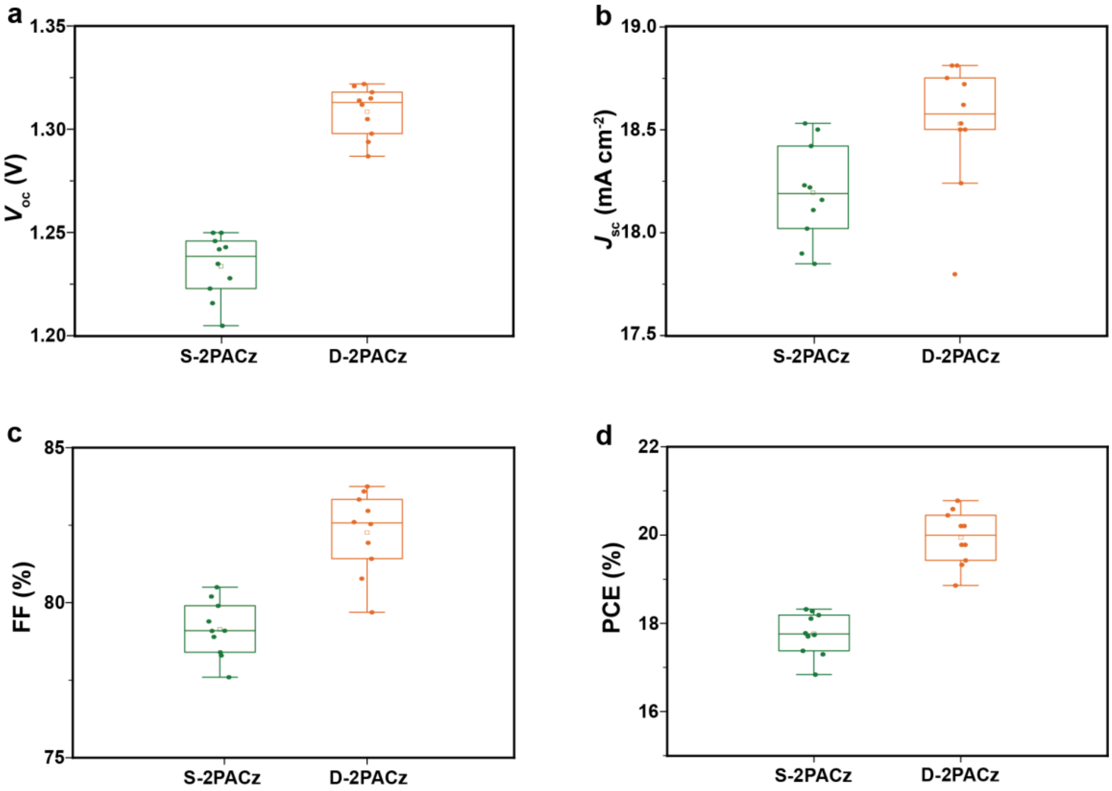
Supplementary Fig. 26 | The third-party certificated performance of D-2P-treated PSC at SIMIT.



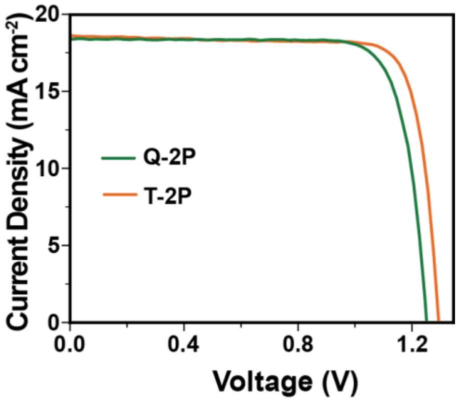
Supplementary Fig. 27 | The EQE of S-2P-treated and D-2P-treated devices.



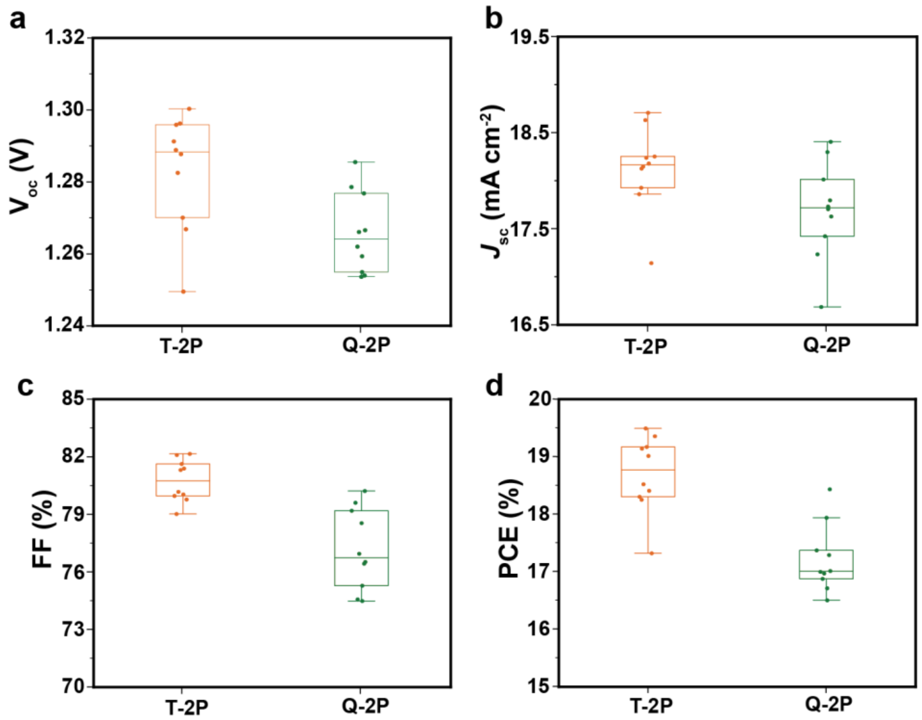
Supplementary Fig. 28 | a Stabilized output power and **b** Forward and reverse scanning of the corresponding device.



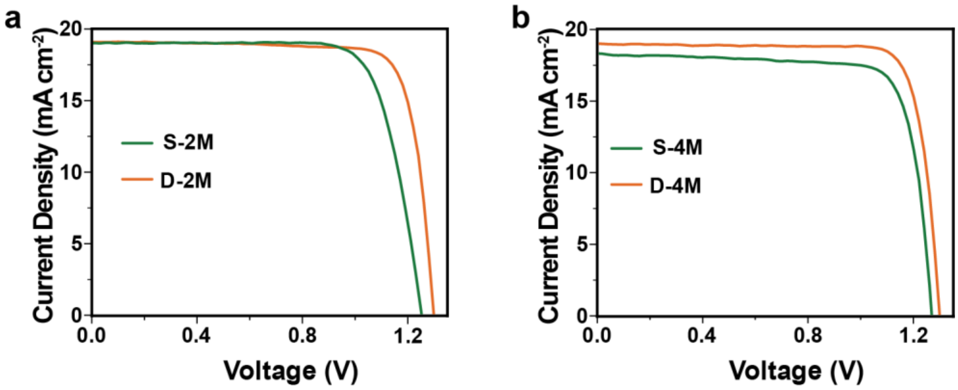
Supplementary Fig. 29 | a V_{oc}, **b** J_{sc}, **c** FF, and **d** PCE distribution of for S-2P-treated and D-2P-treated PSCs.



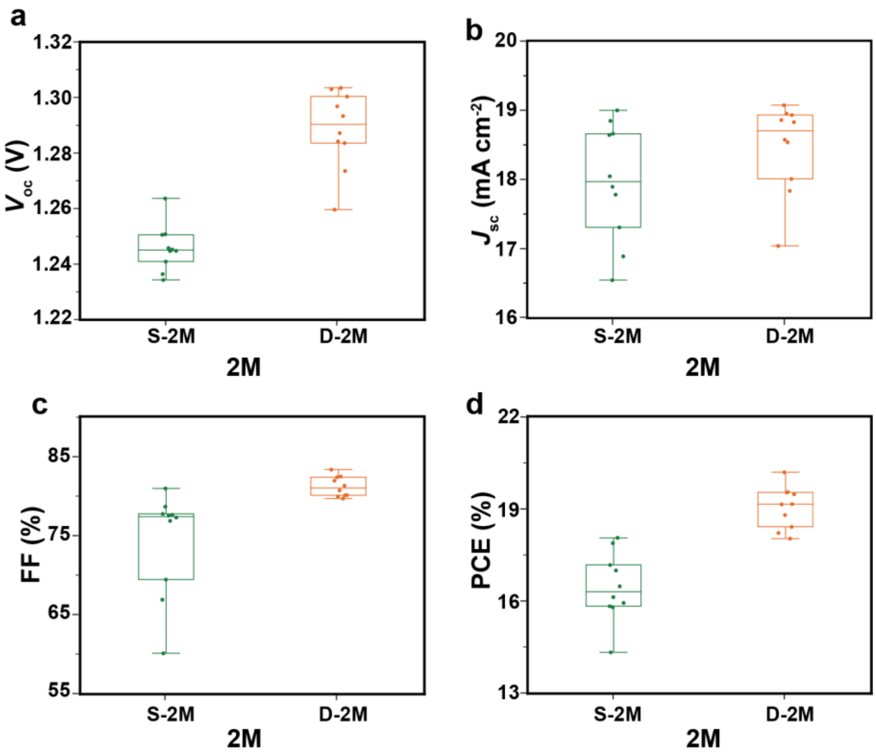
Supplementary Fig. 30 | a *J-V* curves of triple-layer (T-2P)-treated and quadruple-layer (Q-2P)-treated PSCs.



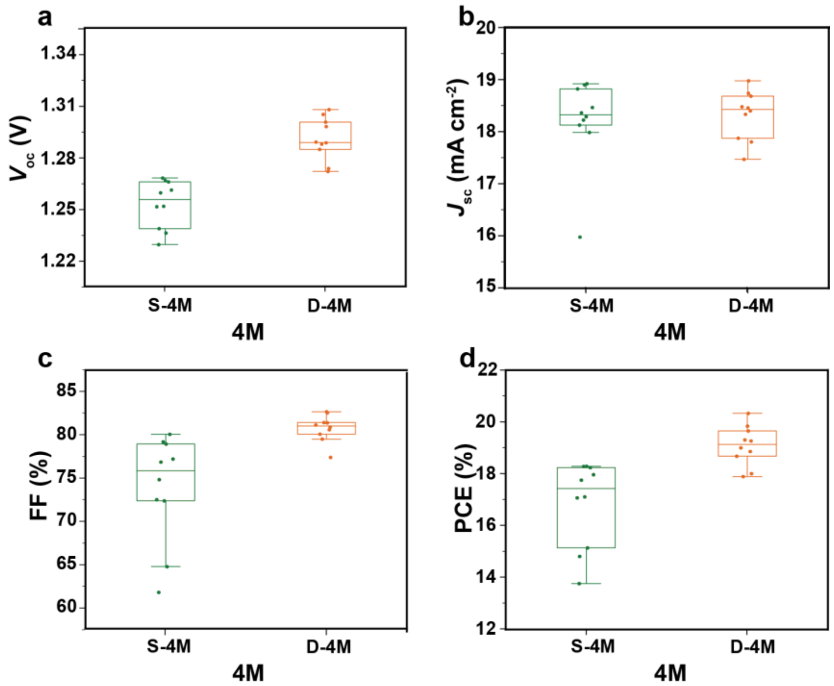
Supplementary Fig. 31 | a V_{oc}, **b** J_{sc}, **c** FF and **d** PCE distribution of triple-layer (T-2P)-treated and quadruple-layer (Q-2P)-treated PSCs.



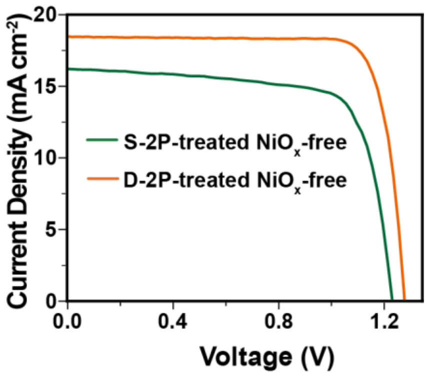
Supplementary Fig. 32 | a *J-V* curves of S-2M-treated and D-2M-treated PSCs. **b** *J-V* curves of S-4M-treated and D-4M-treated PSCs.



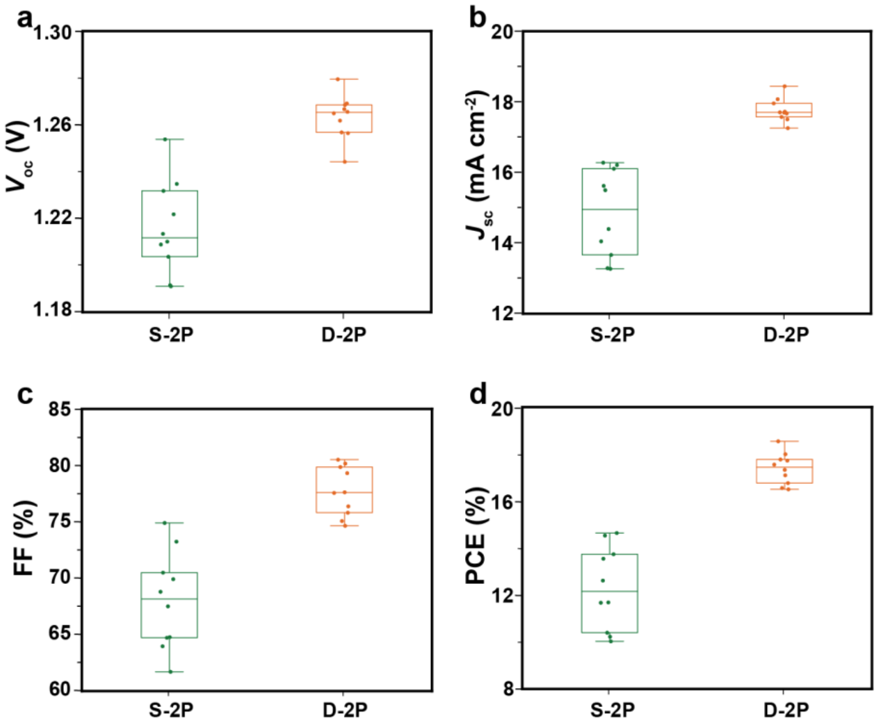
Supplementary Fig. 33 | a V_{oc}, **b** J_{sc}, **c** FF and **d** PCE distribution of S-2M-treated and D-2M-treated PSCs.



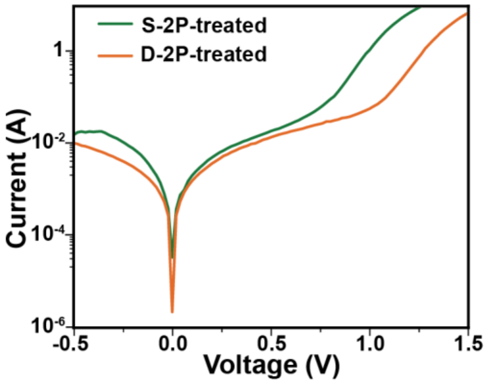
Supplementary Fig. 34 | a V_{oc}, **b** J_{sc}, **c** FF and **d** PCE distribution of S-4M-treated and D-4M-treated PSCs.



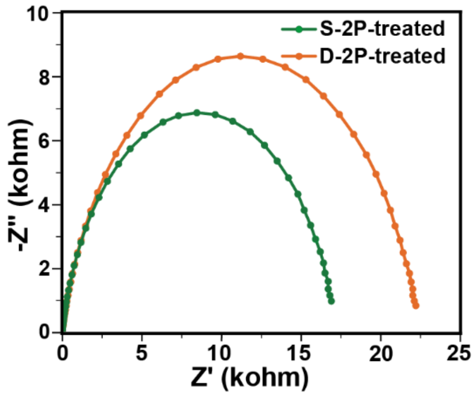
Supplementary Fig. 35 | a *J-V* curves of S-2P-treated and D-2P-treated NiO_x-free PSCs.



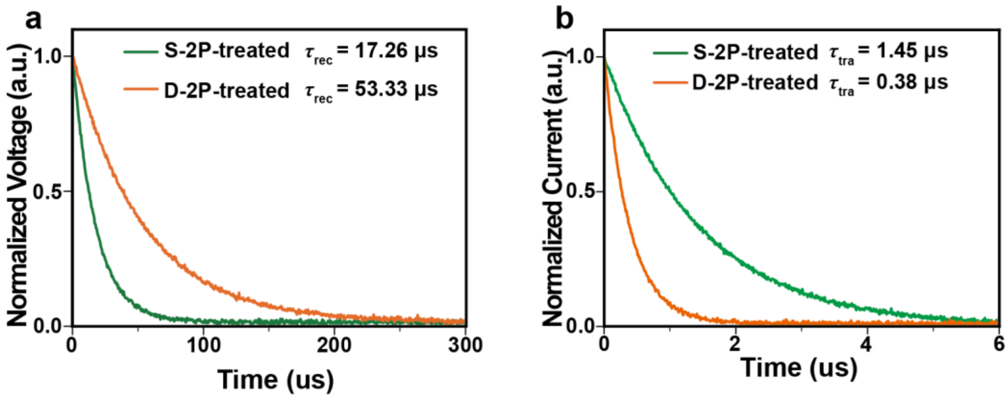
Supplementary Fig. 36 | a V_{oc}, **b** J_{sc}, **c** FF and **d** PCE distribution of S-2P-treated and D-2P-treated NiO_x-free PSCs.



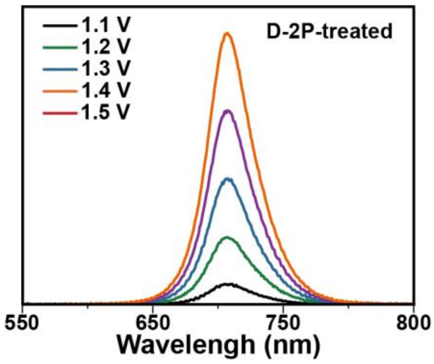
Supplementary Fig. 37 | Dark current of S-2P-treated and D-2P-treated devices.



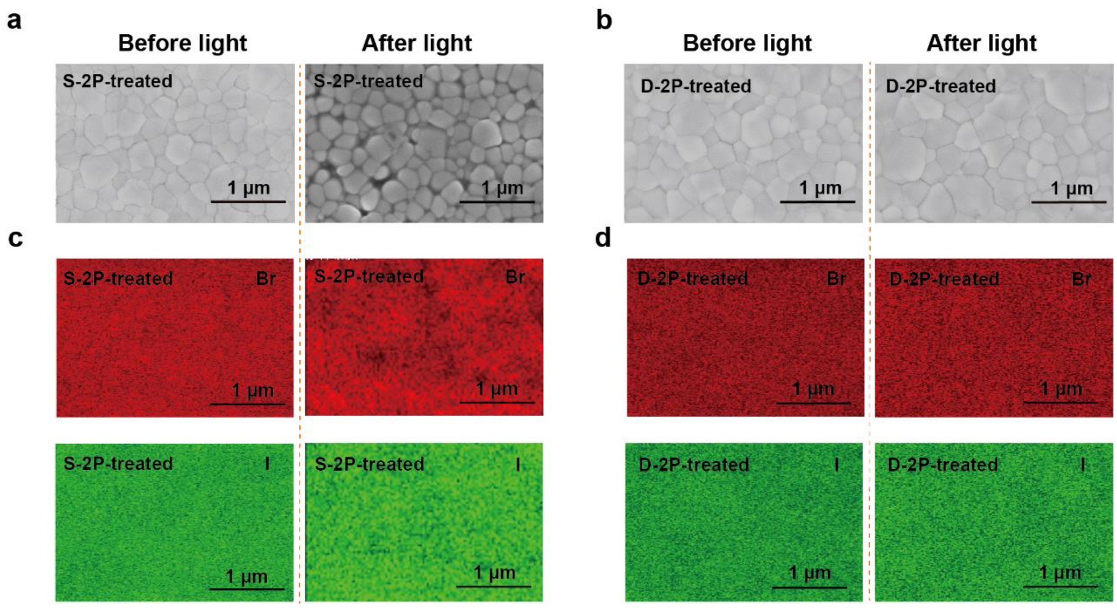
Supplementary Fig. 38 | EIS spectra of S-2P-treated and D-2P-treated devices.



Supplementary Fig. 39 | a S-2P-treated and D-2P-treated devices for **a** TPV and **b** TPC.



Supplementary Fig. 40 | EL spectra of D-2P-treated devices.



Supplementary Fig. 41 | a, **b** The SEM coupled with **c**, **d** EDS mappings of the perovskite films before and after continuous 1-sun illumination at 25 ℃ in N_2 atmosphere.

Supplementary Tables

Supplementary Table 1. Photovoltaic parameters of different concentrations of 2P molecules.

2P (mg ml^{-1})	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)
0.0	1.24	17.68	76.45	16.76
0.5	1.27	18.03	80.81	18.49
1.0	1.29	18.74	78.45	18.88
1.5	1.31	18.77	81.52	19.93
2.0	1.32	18.81	83.59	20.80
2.5	1.31	18.86	81.88	20.02
3.0	1.30	18.83	81.12	19.97

Supplementary Table 2. V_{oc} and PCE of WBG PSCs.

Ref	Perovskite	E_g (eV)	V_{oc} (V)	PCE (%)
1	$FA_{0.83}Cs_{0.17}Pb(I_{0.6}Br_{0.4})_3$	1.74	1.20	17.1
2	$Cs_{0.12}MA_{0.05}FA_{0.83}Pb(I_{0.6}Br_{0.4})_3$	1.74	1.25	19.3
3	$Cs_{0.15}FA_{0.79}MA_{0.06}Pb(I_{0.7}Br_{0.3})_3$	1.74	1.25	20.4
4	$Cs_{0.3}FA_{0.6}DMA_{0.1}Pb(I_{0.7}Br_{0.3})_3$	1.75	1.33	20.3
5	$FA_{0.8}Cs_{0.2}Pb(I_{0.65}Br_{0.35})_3$	1.75	1.22	16.2
6	$FA_{0.8}Cs_{0.2}Pb(I_{0.7}Br_{0.3})_3$	1.75	1.21	20.2
7	$FA_{0.8}Cs_{0.2}Pb(I_{0.7}Br_{0.3})_3$	1.75	1.24	18.2
8	$(FA_{0.6}MA_{0.4})_{0.9}Cs_{0.1}Pb(I_{0.6}Br_{0.4})_3$	1.75	1.26	18.3
9	$FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$	1.75	1.23	17.8
10	$MA_{0.6}FA_{0.4}Pb(I_{0.6}Br_{0.4})_3$	1.76	1.17	14.7
11	$FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$	1.77	1.17	16.5
12	$FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$	1.77	1.28	18.6
13	$FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$	1.77	1.28	17.7
14	$Cs_{0.12}FA_{0.8}MA_{0.08}PbI_{1.8}Br_{1.2}$	1.77	1.29	15.1
15	$FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})_3$	1.77	1.22	16.5
16	$FA_{0.8}Cs_{0.2}Pb(I_{0.6}Br_{0.4})3$	1.77	1.22	17.3
17	$Cs_{0.2}FA_{0.8}Pb(I_{0.6}Br_{0.4})_3$	1.78	1.32	19.6
18	$(FA_{0.8}Cs_{0.2})$ $Pb(I_{0.6}Br0_{.4})_3$	1.78	1.36	19.8
19	$Cs_{0.25}FA_{0.75}Pb(I_{0.6}Br_{0.4})_3$	1.79	1.26	17.8
20	$FA_{0.83}Cs_{0.17}Pb(I_{0.6}Br_{0.4})_3$	1.79	1.22	17.0
21	$Cs_{0.2}FA_{0.8}Pb(I_{0.6}Br_{0.4})_3$	1.79	1.32	20.1
22	$FA_{0.83}Cs_{0.17}Pb(I_{0.52}Br_{0.48})_3$	1.79	1.26	16.5
23	$FA_{0.85}Cs_{0.15}Pb(I_{0.6}Br_{0.4})_3$	1.8	1.27	17.0
24	$FA_{0.6}Cs_{0.4}Pb(I_{0.75}Br_{0.25})_3$	1.8	1.26	17.7
25	$MA_{0.9}FA_{0.1}Pb(I_{0.6}Br_{0.4})_3$	1.81	1.21	17.1
26	$FA_{0.8}Cs_{0.2}Pb(I_{0.5}Br_{0.5})_3$	1.82	1.20	14.1
This Work	$FA_{0.8}Cs_{0.15}MA_{0.05}Pb(I_{0.7}Br_{0.3})_3$	1.75	1.32	20.8

Supplementary Table 3. Photovoltaic parameters of S-2P-treated and D-2P-treated devices. Hysteresis index (HI) was calculated by (PCE_{RS}-PCE_{FS})/PCE_{RS}.

Devices	Direction	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)	*HI* (%)
S-2P-treated	FS	1.25	18.36	76.37	17.62	3.9
	RS	1.25	18.53	79.09	18.33	
D-2P-treated	FS	1.31	18.74	83.53	20.55	1.2
	RS	1.32	18.81	83.59	20.80	

Supplementary Table 4. Photovoltaic parameters of triple-layer (T-2P)-treated and quadruple-layer (Q-2P)-treated PSCs.

Type	Device	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)
2P	T-2P	1.30	18.63	81.31	19.49
	Q-2P	1.25	18.40	80.22	18.43

Supplementary Table 5. Photovoltaic parameters of S-2M-treated, D-2M-treated, S-4M-treated and D-4M-treated PSCs.

Type	Device	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)
2M	S-2M	1.25	18.99	76.22	18.03
	D-2M	1.30	19.07	80.94	20.03
4M	S-4M	1.26	18.29	78.93	18.29
	D-4M	1.29	18.97	82.56	20.33

Supplementary Table 6. Photovoltaic parameters of S-2P-treated and D-2P-treated NiO_x-free PSCs.

Type	Device	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)
NiO_x-free	S-2P	1.23	16.20	73.24	14.63
	D-2P	1.28	18.07	80.50	18.54

Supplementary Table 7. The photovoltaic parameters of a 4-T PTSCs.

Device	V_{oc} (V)	J_{sc} (mA cm^{-2})	FF (%)	PCE (%)
ST-WBG PSCs	1.30	18.02	82.43	19.35
Single-junction NBG PSCs	0.86	31.52	80.07	21.7
Filtered NBG PSCs	0.84	12.87	80.31	8.73
4T tandem cell				28.08

Supplementary Methods

Characterization of perovskite single crystals and films

UV-Visible absorption spectra of perovskite films were carried out on a Lambda 950 UV-vis spectrophotometer.

Ultraviolet photoelectron spectra (UPS). The equipment model is Thermo Scientific ESCA Lab 250Xi. Perovskite films are measured in an ultra-high-vacuum (UHV) chamber (base pressure below 5×10^{-9} mbar). Au as the standard sample for calibration. He I (hν = 21.22 eV) emission source was employed; the photon line width was about 250 meV and the minimum spot size about 1 mm. The work function (W_F) of perovskite films can be calculated from following equation:

$$W_{\mathrm{F}} = hv - (E_{\mathrm{Ferni}} - E_{\mathrm{Cutoff}}) \quad (1)$$

where the E_{cutoff} is the steep edge position in the UPS spectra. The valence band maximum (E_{VBM}) values of the films can be calculated by the equation:

$$E_{VBM} = W_{\mathrm{F}} + VB \quad (2)$$

The optical bandgap (E_g) of perovskite films can be obtained by measuring their absorption spectra. Thus, the conduction band minimum (E_{CBM}) values of the films can be calculated by the equation:

$$E_{CBM} = E_{VBM} - E_{\mathrm{g}} \quad (3)$$

The Fourier transform infrared spectroscopy (FTIR) was measured on a PerkinElmer Spectrum Two infrared spectrometer.

Scanning electron microscopy (SEM). The top-view and cross-section SEM images of perovskite films were characterized by field-emission Scanning Electron Microscopy (SEM; JSM-7500F, JEOL).

The external quantum efficiency (EQE) spectra were obtained by using a QE system (EnliTech).

TEM sample preparation. The sectional specimens were prepared from the PSCs using focused ion beam (FIB) cutting. Before the FIB cutting, a Pt/carbon layer was deposited onto the Ag electrode of the sample to avoid the milling damage. All as-prepared specimens were welded on the Cu grid and then were transferred into TEM setup within 30 s.

Kelvin probe force microscopy (KPFM) was operated combined with an atomic force microscopy (AFM, MFP-3D, OXFORD, Britain) and a HF2LI Lock-in amplifier (Zurich Instruments).

Confocal fluorescence microscopy (TCFM) images were performed using a custom sample scanning confocal microscope with TCSPC (PicoHarp 300) capabilities built around a Nikon TE-2000 inverted microscope fitted with an infinity corrected × 50 dry objective (Nikon L Plan, NA 0.7, CC 0-1.2). A 470-nm pulsed diode laser (PDL-800 LDH-P-C-470B, 350 ps pulse width) was used for excitation with a repetition rate of 1 MHz for time-resolved PL measurements and 40 MHz for collecting fluorescence images. A 532-nm CW laser (CrystaLaser, GCL532-005-L) was used to excite local regions and monitor the PL rises over time. Samples were excited face-on through the flow cell's coverslip and the emission was filtered through a 700-nm long passage filter then directed to a Micro Photon Devices (MPD) PDM Series single photon avalanche photodiode with a 50-µm active area. The sample stage was controlled using a piezo controller (Physik Instrumente E-710) with custom software.

Grazing incident X-ray diffraction (GIXRD) measurement. We consider the perovskite films as quasi-isotropic. According to the Bragg's Law and generalized Hooke's Law[27], the 2θ-$\sin^2\varphi$ can be given by the equation,

$$\sigma = -\frac{E}{2(1+\nu)}\frac{\pi}{180°}cot\theta_0\frac{\partial(2\theta)}{\partial \sin^2\varphi} \qquad (4)$$

where E is the perovskite modulus (10 GPa), v is Poisson's ratio of the perovskite (0.3), θ_0 is half of the scattering angle $2\theta_0$, corresponding to a given diffraction peak for strain-free perovskite[28]. 2θ is the scattering angle for the actual perovskite. φ is the angle the diffraction vector N_k with respect to the sample surface normal. By fitting the 2θ as a function of $\sin^2\varphi$, the perovskite film strain can be calculated from the above formula, and the slope of the fitted line stands for the scale of the residual strain.

Photoluminescence (PL) spectra. The steady-state PL spectra were measured by fluorescence spectrophotometer (Edinburgh instruments FLS 980) provided by a xenon lamp with an intensity of 100 mW cm^{-2}. ITO/NiO_x/S-2P/Perovskite and ITO/NiO_x/D-2P/Perovskite films were excited at 475 nm. The excitation source for time-resolved photoluminescence (TRPL) spectra was a laser diode (450 nm, 2 nJ cm^{-2}).

Thermal admittance spectroscopy (TAS) analysis was obtained by capacitance-frequency curve using PARSTAT 4000 was used to measure the capacitance-frequency curve. The energy profile of trap density of states could be fitted from the equation:

$$N_t(E_\omega) = -\frac{V_{bi}}{eW}\frac{dC}{d\omega}\frac{\omega}{k_B T} \quad (5)$$

Where V_{bi} is the built in voltage, ω is angular frequency, e is the elementary charge, W is the depletion width, C is the capacitance, k_B is the Boltzmann constant and T is the temperature.

Device characterization

Electrochemical Impedance Spectroscopy (EIS) were measured by the electrochemical workstation. The light source was simulated by a Xenon-lamp-based solar simulator (Enli Tech), and the illumination intensity was tuned by filters. The impedance spectra were recorded by the E4980A Precision LCR Meter from Agilent with homemade software in the frequency range of 2000 to 0.2 kHz. For transport resistance (R_{ct}) and recombination resistance (R_{rec}) measurements, the devices ware applied with a bias voltage equaled to V_{oc} under dark conditions.

Transient photocurrent (TPC) and transient photovoltage (TPV) measurement were measured on a home-made system. A white light bias is generated from an array of diodes (100 mW cm^{-2}) and red-light pulse diodes (0.05 s square pulse width, 100 ns rise and fall time, 5 mW cm^{-2}) controlled by a fast solid–state switch were used as the perturbation source. TPC was measured using 20 Ω external series resistance to operate the device in short-circuit. Similarly, TPV was measured using 1 MΩ external series resistance to operate the device in open circuit. The voltage dynamics on the resistors were recorded on a digital oscilloscope (Tektronix MDO3032). The perturbation red light source was set to a suitably low level to the white diodes array with light intensity equivalent to 100 mW cm^{-2} of a standard solar simulator.

Mott-Schottky measurement can determine the type of semiconductor, the charge carrier concentration, and the flat band potential. The space charge layer capacitance (C_{SC}) and potential E can be described by Mott-Schottky equation. The Mott-Schottky equation is a mathematical equation that describes the differential (C_{SC}) capacitance of the space charge layer of a semiconductor as a function of the surface potential of the semiconductor (E_{fb}). Mott-Schottky formula:

$$C_{SC}^{-2} = \frac{2}{\varepsilon\varepsilon_0 e N_D}(E - E_{fb} - \frac{kT}{e}) \quad (6)$$

$$C_{SC}^{-2} = \frac{2}{\varepsilon\varepsilon_0 e N_A}(E - E_{fb} - \frac{kT}{e}) \quad (7)$$

In the equation, E_{fb} was the flat band potential; N_D and N_A carrier concentrations; ε is the relative dielectric constant; ε_0 is the vacuum permittivity; k is the Boltzmann constant; T is the absolute temperature; e is the electric quantity per unit charge. An AC disturbance signal with fixed amplitude and fixed frequency was superimposed at different DC potentials, and the impedance values at different potentials were measured. The capacitance in the depletion zone was calculated according to the imaginary part of the impedance:

$$C_{SC} = -\frac{1}{2\pi f Z''} \quad (8)$$

In the equation: Z'' is the imaginary part of impedance; f is the sine wave frequency.

Temperature-dependent conductivity measurements for E_a extraction. E_a can be extracted from the Nernst-Einstein equation:

$$ln\sigma T = ln\sigma_0 - E_a(\frac{1}{k_B T}) \quad (9)$$

where σ_0 is a constant, k_B is the Boltzmann constant, and T is temperature.

Theoretical Calculations

Intermolecular interaction. The geometry optimization and electronic structure calculation were carried out using the Material studio (MS). A uniform 2 × 2 × 1 Monkhorst-Pack k-point mesh and a plane-wave energy cutoff of 400 eV were adopted to the structural optimization. We used the Perdew-Burke-Ernzerhof (PBE) functional[29] to express the exchange-correlation interactions and projector-augmented wave (PAW) method[30] to account for the electron-ion interaction. The van der Waals interactions was described by the Grimme DFT-D3 correction[31].

Formation energy of perovskite. We constructed 12 formula unit (f.u.) supercells of $FAPbI_3$ and $FAPbBr_3$. The formation energies of $FAPbI_3$ and $FAPbBr_3$ were calculated with respect to the precursor phases. We chose FAI, FABr, PbI_2 and $PbBr_2$ as precursors. The formation energies can be calculated by the equations,

$$E_{\mathrm{FAPbI_3}} = E[\mathrm{FAPbI_3}] - E[\mathrm{FAI}] - E[\mathrm{PbI_2}] \tag{10}$$

$$E_{\mathrm{FAPbBr_3}} = E[\mathrm{FAPbBr_3}] - E[\mathrm{FABr}] - E[\mathrm{PbBr_2}] \tag{11}$$

Here, E is the total energy of the respective fully relaxed phases calculated with DFT. All calculations were performed at 0 K, without considering thermal effects. We performed variable cell Kohn-Sham DFT calculations of all structures within the Generalized Gradient Approximation (GGA) in the form of the Perdew-Burke-Ernzerhof (PBE) exchange correlation functional with Grimme D3 dispersion corrections. We used Quantum Espresso with ultra-soft pseudo-potentials for valence-core electron interactions and a plane wave basis set with a 60 Ry kinetic energy cutoff and a 480 Ry density cutoff[32].

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